

2025 University Rover Challenge: System Acceptance Review Report

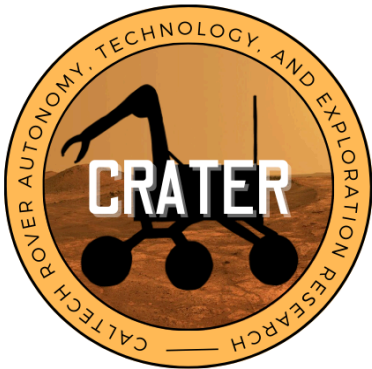
Team: Caltech Rover Autonomy, Technology, and Exploration Research (CRATER)

Student Team Lead

Ji Hyun Jeon
California Institute of Technology
Undergraduate, Computer Science
jhjeon@caltech.edu

Faculty Advisor

Dr. Soon-Jo Chung
California Institute of Technology
Professor of Aerospace Eng. and Control and Dynamical Systems
Jet Propulsion Laboratory Senior Research Scientist
sichung@caltech.edu



Caltech

University Rover Challenge

Introduction

CRATER was founded in September 2022 by Caltech undergraduate students who joined with the shared goal of creating a rover. Since our founding, we overcame various hurdles and pushed through our development cycle to create two fully working chassis, go through three design iterations of the arm, and compete onsite in two consecutive years of CIRC (Canadian International Rover Challenge) in August 2023 and 2024. For the past 6 months starting in September 2024, we have focused our work on getting our rover ready for URC. We have implemented full wireless driving capabilities, confirmed the mechanical capabilities of our arm and developed joint based control for the arm.

Core Rover Systems

Mechanical Systems

The mechanical subteam designed the rover's chassis, suspension, and transmission. We implemented a rocker-bogie suspension for stable terrain traversal, with a differential bar to enhance stability. To balance durability and speed, we used aluminum linkages, chassis, and a differential bar, further reducing mass with an isogrid pattern on rover plates.

Our chain-and-sprocket transmission ensures high torque and speed while positioning motors safely above the wheels. Maxon EC 60 brushless motors with 26:1 planetary gearheads provide 13 Nm output torque per motor. The chassis houses electrical, science, and manipulator systems, and has multiple removable plates to allow for easy access of internal systems. Wheel's inflatable polyurethane wheels improve traction, clearance around obstacles, and skid-steer capability.

Manipulator

The 6-DOF robotic arm is designed to lift 5 kg, reach 1.5m, and fold into a 0.4m x 0.4m x 0.2m space for competition constraints. Its hollow aluminum linkages connect a base joint, four intermediate joints, and the end effector. The base joint uses a Nema 34 stepper motor. The intermediate joints use a brushless dc motor coupled with a non-backdrivable 100:1 gearbox, ensuring sufficient torque and precision. The end effector uses a lead screw to grip objects with fine movement, adapting to any orientation and ensuring secure parallel grip. The arm is controlled via ROS2 commands over CAN bus to ODrive motor controllers, with inverse kinematics determining joint angles for precise motion.

Electrical and Power Systems

This subsystem handles onboard electronics and power distribution, ensuring reliable power and signals for the rover's numerous actuators and sensors. We prioritized component interchangeability for easy repair and replacement by using a modular network of off-the-shelf power converters and motor

controllers, mounted on an easily modifiable acrylic sheet. Strategic connector placement enhances modularity and maintenance. For efficiency and safety, we implemented an active balancing BMS that ensures uniform cell discharge and provides real-time data via Bluetooth or RS485. A single-point grounding scheme improves signal robustness against EMI.

Communication Systems

The rover employs three communication lines: a 1.3 GHz video feed, a 2.4 GHz WiFi link for primary command transmission, and a 900 MHz radio link as a backup. At CIRC 2024, we achieved stable video and command transmission over WiFi but are integrating a 1.3 GHz video feed for higher resolution and speed.

1.3 GHz Video – Uses a Matek transmitter/receiver with a 600 TVL FPV camera. An EasyCam cable converts the analog signal for Raspberry Pi processing.

2.4 GHz WiFi – Utilizes Ubiquiti Rocket 2AC units, with a sector antenna at the base and an omnidirectional antenna on the rover for high-speed, low-latency short-range communication.

900 MHz Radio – Adafruit Feathers with LoRa chips provide long-range communication, meeting extreme delivery challenge requirements. A 15dbi Yagi antenna extends the range to 2 km.

Software

The rover and base station use Robot Operating System 2 (ROS) Control as its framework for modularity and reliability of communication packets. Modules are written in Python on the base station for ease of prototyping and C++ for the on-board computer for more efficient processing. This is used alongside an Xbox remote controller to drive the wheels and arm motors, all running on ODrive motor controllers. We have leveraged existing ROS packages to handle communication between the onboard computer and our microcontrollers, actuators and sensors. Both our Jetson Orin Nano and Raspberry Pi 5 are programmed with such libraries, and communicate with the on-board computer through the ROS control framework. For camera feeds, we leverage WebRTC to send low-bandwidth live video feeds, and PointCloud libraries for onboard processing of the RealSense camera.

For autonomy, we leverage the RealSense drive camera and an arm-based camera alongside a gps module and wheel encoders, ensuring we have a consistent understanding of the surrounding environment. These sensors will also aid the driver during manual control, enabling teleoperation.

Approach to Competition Missions

Delivery Mission

We will drive the rover remotely through a handheld Xbox controller. We will communicate with the rover through our 2.4 GHz WiFi link (and through our 900 MHz link for rover commands if the 2.4 GHz

link fails). We will navigate using an FPV camera attached near the end-effector of the manipulator as well as with an on-rover GPS. Our rocker-bogie system has proven successful at traversing rough desert terrain, allowing us to angle the rover or drive it wherever we see fit. We will angle our 6-DOF manipulator to pick up objects and use the manipulator camera to tell when we can close our gripper and lift the object.

Equipment Servicing Mission

The manipulator was designed to be able to complete the tasks of the Equipment Servicing Mission. The end effector can reliably grip on objects up to 10 cm in dimensions and move accurately in task space. A tip attached to the linear actuator can be attached to one of the claws of the gripper for typing on a keyboard. The end-effector camera will allow for easier view of the task field and the computer display.

Autonomous Navigation Mission

We will hold the manipulator end-effector in a fixed position to keep the camera always facing in front of the rover. Using the GPS, the rover will orient itself to point towards its next target, and drive in a straight line to there. When within a 10-foot radius of its target coordinates, the rover will stop and send a message over the WiFi link to the operator that it has reached its destination.

Snapshots of the video stream will be processed every few seconds for object detection, and should an obstacle be found, the rover will start turning right until said obstacle is out of sight. Then it will drive forward for a few seconds and start turning right again, the process repeating until the obstacle has been passed and the rover can make a beeline for its target again.

Testing and Operations

Testing To-Date

DRIVE CAPABILITIES: We have tested our rover both on rocky desert terrain and on small hills around campus. The rover can adequately climb hills of up to 3 feet in height and can power through desert brush and ditches about half its height.

MANIPULATOR CAPABILITIES: We have confirmed that the manipulator can lift a 5 kg payload with maximum arm extension under manual joint control. We have also tested the end effector's gripping capabilities with a variety of objects, including irregularly shaped and delicate items

REMOTE CONTROL: In the desert and on campus, we have controlled the rover remotely over the 2.4GHz system from up to 1 km away, which gave a spotty connection on campus but worked reliably in the remote desert.

POWER CONSUMPTION: Power draw of components such as the manipulator actuators and drive motors have been analyzed as well. Without the manipulator, the entire rover never drew above 10A, the majority of the power draw coming from the drive motors. We estimate that with all of the manipulator motors going, we will draw a max of 20A.

Component/Subsystem Testing

VIDEO FEED: Our new FPV camera, replacing the previous RealSense USB digital camera, has been successfully tested in-lab for clarity and close-range precision, demonstrating the ability to read text from a few inches away. The next phase of testing involves verifying the video feed's reliability over long distances by transmitting video across campus. This will allow us to evaluate signal integrity and latency when using the 1.3 GHz transmitter, ensuring smooth video streaming for remote operation. Additionally, we will assess the performance of controlling the rover solely off the 900 MHz link across campus to verify robustness in desert terrain.

MANIPULATOR: Upcoming tests will focus on evaluating the full range of motion across all six degrees of freedom to ensure smooth operation in real-world scenarios. Additional testing will involve joint precision and stability under dynamic conditions, ensuring that the arm maintains accuracy while lifting and manipulating objects. The end effector capabilities will be further tested, especially of the linear actuator attachment used for typing on a keyboard

System and Operations Testing

OBJECT MANIPULATION: To test the manipulator's effectiveness in competition scenarios, we will conduct trials simulating the Equipment Servicing Mission tasks. This includes picking up objects from different surfaces, interacting with control panels, and using the gripper for fine movements like pressing buttons and flipping switches. We will also evaluate the efficiency of inverse kinematics algorithms by executing pre-programmed sequences to determine whether the arm reaches target positions smoothly and accurately. For autonomy, we will test that our OpenCV based program is capable of getting accurate coordinates of the necessary keys and that the arm is able to precisely execute the operation.

AUTONOMY: For the Autonomous Navigation Mission, we will conduct structured tests focusing on GPS waypoint tracking, obstacle detection, and path planning. The rover will be programmed to travel between waypoints in an open outdoor environment, using its front-facing manipulator-mounted camera to detect and avoid obstacles. We will systematically introduce various obstacles—such as large rocks, barriers, and ditches—to evaluate the effectiveness of the object detection pipeline using OpenCV and PointCloud libraries. Additionally, we will assess the autonomous rerouting logic, ensuring the rover correctly bypasses obstacles while staying on course. Communication reliability during autonomy will also be tested to confirm the rover can send status updates to the operator when it reaches designated waypoints.

Budget Table

Income

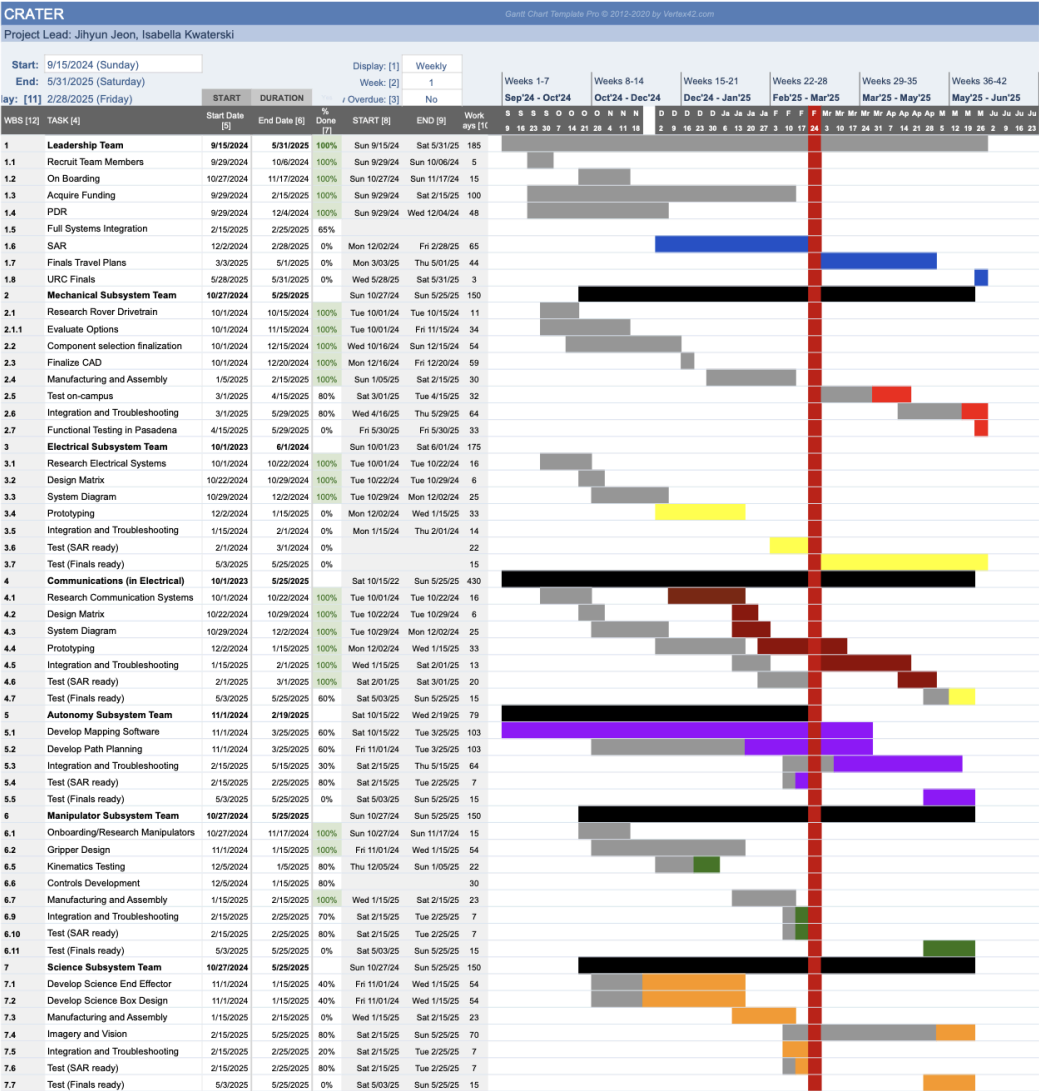
Source	Amount Estimate	Description
Caltech Housner Fund	\$21,000	Funds received after submitting proposal to the Caltech Housner Fund
California Space Grant Consortium	\$2,500	Funds received after applying for CaSGC award
AIAA Funds	\$30,000	Leftover Club Funds
Corporate Sponsorships	\$2,000	Protocase sponsorship through URC
Total	\$55,500	

Expenses

Category	Cost
Current Final Rover and Base Station Hardware	\$17,353
Prototyping, Spare Parts & Other (Estimate)	\$17,353
Testing Travel Costs	\$1,200
URC 2025 Travel Costs	\$13,464
Total	\$49,370

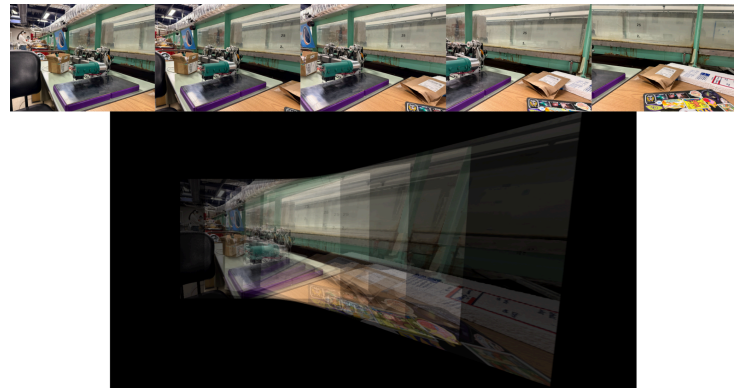
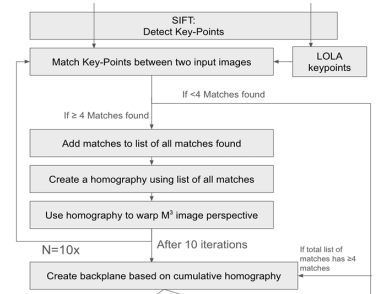
We have raised sufficient funding since the PDR for all rover and prototyping hardware and machining costs. We will pursue additional funding from the Caltech Housner Fund’s Spring Term application cycle to travel to the Mars Desert Research Station should we be selected to compete in the URC 2025 Finals.

Gantt Chart



The rover's science subsystem this year focuses on rock profiles and analysis, with testing procedures largely carried over from prior years. Sampling will be entirely manned, with the rover driver navigating to sampling locations.

This year's task requires imaging of rock surfaces and strata. We will take a wide angle panorama at each site, consisting of a video taken by moving the arm linearly across 30cm directly in front of the rover, parallel with the front face of the rover. During this video, motion will be at constant velocity, allowing for post-processing to mesh frames together. This post-processing relies heavily on an iterative approach to the SIFT algorithm as described in Gauld et. al¹ (2025). A flowchart of this method is shown to the right, modified to mesh multiple images one after the other. This is done to various consecutive frames of the video, allowing for a composite image to be gathered. We also save the images individually for post-panorama correction



Sample locations will be given to the driver of the rover, who will drive in front of the sample such that it can be reached by the manipulator arm while not interfering with the sample itself. There are two forms of samples we are considering: loose sediment/soil and rock monoliths. All sampling will be done using the end effector attached to the manipulator arm. For soil samples, samples are taken using a hot-swappable scooping claw on the end of the manipulator arm. The manipulator arm is then positioned over a rotating chamber enclosure on the chassis into which the soil is deposited². Once deposited into the rover, the sample is added to water and sonicated to extract the biomaterials from the soil. The resultant mixture is then filtered, with the filtrate used for testing. To collect and cache a sample, the rover can drill to depths up to 15 cm, and contain it within our custom auger. After drilling, a probe will descend to our sample depth and record soil temperature and humidity.

The first metric we use looks for extant life in the soil sample. We look for biomaterials that degrade over time, primarily proteins and amino acids. We test this with the Bradford protein assay, adding Coomassie brilliant blue G-250 to the sample and checking absorption. With an increase in protein concentration, the absorption will increase, as measured by an onboard spectrophotometer³⁴. We will also look for extinct life in the sample. We look for kerogen sources, a natural oil present due to organic material degradation. For this, we will use raman spectrometry onboard the rover to detect emission peaks for signals of life. Our design is heavily inspired by OpenRAMAN but is modified to live on the rover body. Our test metrics are solidified for the rover. Our future steps are primarily integration, implementing mechanisms for our test selections and integrating into the rover chassis.

¹ Gauld 2025 *Image-Based Localization for ... Images ...* (LPSC 2025)

² Robe 2003 *Extraction of DNA from Soil*

³ Bradford 1975 *A Rapid and Sensitive Method ... Utilizing the Principle of Protein-Dye Binding*

⁴ Simoni 2002 *Benedict's Solution, a Reagent for Measuring Reducing Sugars*